

Multiple Choice

1. **Answer:** D

Options: A) 42; B) 144; C) 156; D) 132; E) 96

Note: Correct option: D - 132

2. **Answer:** A

Options: A) 7.0; B) 2.0; C) 4.5; D) 1.0; E) 5.0

Note: Correct option: A - 7.0

3. **Answer:** A

Options: A) 2.0; B) 3.25; C) 34.667; D) 7.5; E) 6.125

Note: Correct option: A - 2.0

4. **Answer:** A

Options: A) 12000000; B) 5000000; C) 2000000; D) 40320; E) 7257600

Note: Correct option: A - 12000000

5. **Answer:** C

Options: A) 9; B) 8; C) 0; D) 1; E) 6

Note: Correct option: C - 0

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '\$6049'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $\frac{8}{45}$ '.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '392'.

Long Form Response

11. **Answer:** There are two cases, one where the AB is the base, and the other where AB is a leg. For the case where AB is the base, we can create the third point C anywhere on the line perpendicular to AB at the midpoint of AB . There are 4 points on that line. For the case where AB is a leg, since AB is two units, we can create a point C two units away from either A or B . There are two such points. In total, there are $2 + 4 = 6$. [asy] draw((0,0)--(0,6)--(6,6)--(6,0)--cycle,linewidth(1)); for(int i=1;i<6;++i) {for(int j=1;j<6;++j) {dot((i,j));} } draw((2,2)--(4,2),linewidth(1)); label("A",(2,2),SW); label("B",(4,2),SE); label("C",(3,1), SE); label("C",(3,3), SE); label("C",(3,4), SE); label("C",(3,5), SE); label("C",(4,4), SE); label("C",(2,4), SE); [/asy]
Note: Students should show all steps in their mathematical solution.
12. **Answer:** $\frac{3}{10}$. Explain why this is the correct answer and provide supporting evidence. Show the calculation or reasoning that leads to this conclusion.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.